



SPX-XSP Options Visualizer

[Quick Start Guide](#)

Get up and running in minutes

Part 1 — Setting Up a Strategy with Manual Prices

This section shows how to set up a two-instrument strategy and view the combined risk graph using manually entered prices. No broker connection is needed.

Step 1 — Select the SPX Strategy

In the **Instrument 1 (SPX)** section, select the direction (**Debit** or **Credit**) and the strategy type. In this example **CS** (Call Spread) is selected with **Debit** direction, giving a Call Debit Spread. Enter the **Long Call** strike, the **Width** (distance between strikes), and the Short strike is calculated automatically.

The screenshot shows the 'Instrument 1 (SPX)' configuration window. At the top, 'Debit' is selected in a dropdown menu, and 'CS' (Call Spread) is selected among radio button options. Below, 'Symbol' is 'SPX', 'Qty' is '1', and 'BPR' is '\$265'. The 'Call Debit Spread' section shows 'Long Call' at '7410', 'Width' at '15', and 'Short' at '7425'. A price field on the right shows '2.65' with a padlock icon and '(Db)'.

Note: The prices you enter in the following steps should be obtained from your broker platform (e.g. TWS, Tastytrade) — use the current mid price of the spread or individual legs as shown in your broker's option chain.

Step 2 — Enter the SPX Strategy Price

Type the debit or credit price in the price field (right side of the spread row). The padlock icon next to the field will close, indicating the price is **locked** and will not be overwritten by a broker fetch. The BPR (Buying Power Reduction) updates automatically.

This screenshot is identical to the previous one, but the price field '2.65' is now circled in yellow. The padlock icon next to it is closed, indicating the price is locked. The BPR has updated to '\$290'.

Step 3 — Select the XSP Strategy

In the **Instrument 2 (XSP)** section, select the strategy type. Here **PS** (Put Spread) is selected with **Debit** direction, giving a Put Debit Spread. Enter the **Long Put** strike, the **Width**, and adjust the **Qty** (number of contracts). The Short strike is calculated automatically.

Instrument 2 (XSP)

Debit ☐ None ☐ Custom

☐ Call ☐ Put ☐ CS ☒ PS ☐ CButt ☐ PButt ☐ IC ☐ CCond ☐ PCond ☐ Straddle ☐ Strangle

Symbol: XSP Qty: 5 BPR: \$220

Put Debit Spread

Long Put: 741 Width: 2 Short: 739 0.44 (Db)

Step 4 — Enter the XSP Strategy Price

Type the price for the XSP spread. The padlock closes to lock it. Notice the BPR reflects the total for the quantity entered.

Instrument 2 (XSP)

Debit ☐ None ☐ Custom

☐ Call ☐ Put ☐ CS ☒ PS ☐ CButt ☐ PButt ☐ IC ☐ CCond ☐ PCond ☐ Straddle ☐ Strangle

Symbol: XSP Qty: 5 BPR: \$175

Put Debit Spread

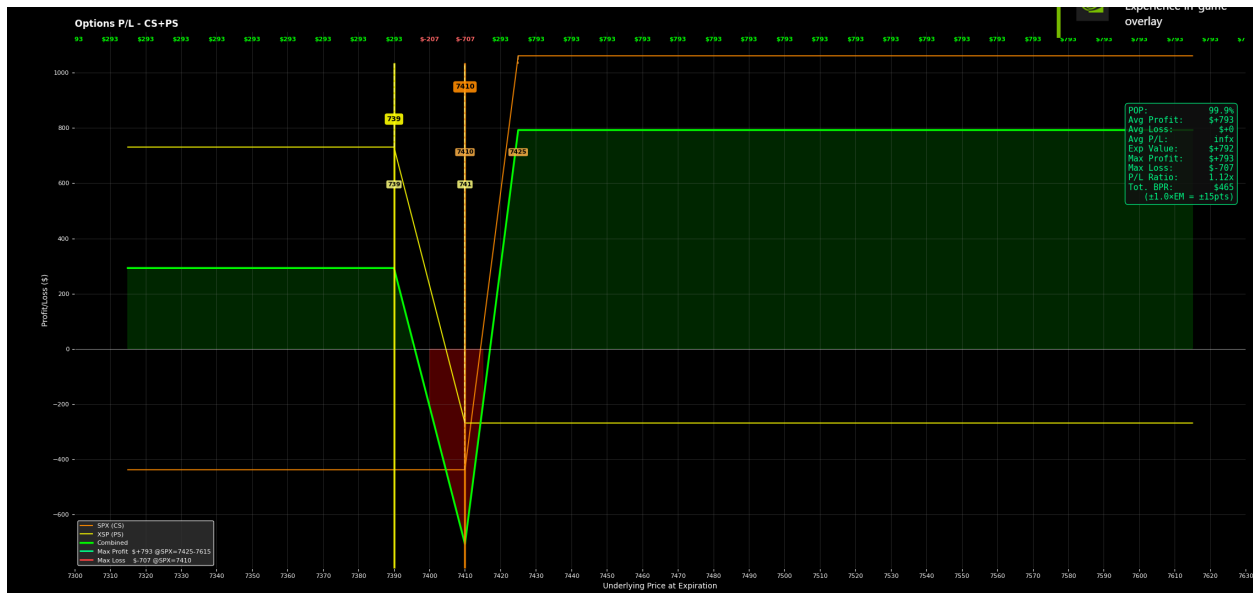
Long Put: 741 Width: 2 Short: 739 0.35 (Db)

Step 5 — View the Risk Graph

The chart updates automatically after each change. It shows:

The orange curve is the **SPX Call Debit Spread**, the yellow curve is the **XSP Put Debit Spread**, and the green curve is the **Combined P/L** at expiration across the full price range.

The statistics box (top right) shows POP, Max Profit, Max Loss, P/L Ratio, and total BPR. Drag the yellow strike flags on the chart to adjust strikes visually.



The examples shown are for illustrative purposes only. The actual graph will vary based on real-time prices. The P/L shown is not representative of real results.

Adding an Additional Hedge

The Additional Hedge section lets you add a third independent position on XSP to further shape the combined P/L profile.

Step 6 — Enable the Additional Hedge

Check the **Enable Additional Hedge** checkbox in the XSP Hedge Options section. Select the hedge strategy type — here **PB** (Put Butterfly) is chosen with **Debit** direction. Set the **Width** (distance between the butterfly legs).

The screenshot shows the 'XSP Hedge Options' dialog box. The 'Enable Additional Hedge' checkbox is checked. The 'Qty' is set to 1. The 'Hedge' dropdown is set to 'Debit'. The 'None' radio button is selected. The 'Call', 'Put', 'CS', 'PS', 'CButt', 'PB', 'IC', 'CCond', 'PCond', 'Straddle', and 'Strangle' radio buttons are visible, with 'PB' being the selected strategy. The 'Debit Put Butterfly' section shows 'Lower' and 'Upper' fields as empty, 'Center' as empty, 'W' as 5, and a lock icon next to a field containing 0.5, with '(Db)' next to it.

Step 7 — Enter the Hedge Center Strike

Enter the **Center** strike for the butterfly. The Lower and Upper strikes are calculated automatically from the center and width. The hedge BPR is shown next to the Enable checkbox.

The screenshot shows the 'XSP Hedge Options' dialog box. The 'Enable Additional Hedge' checkbox is checked. The 'Qty' is set to 1. The 'BPR' is \$50. The 'Hedge' dropdown is set to 'Debit'. The 'None' radio button is selected. The 'Call', 'Put', 'CS', 'PS', 'CButt', 'PB', 'IC', 'CCond', 'PCond', 'Straddle', and 'Strangle' radio buttons are visible, with 'PB' being the selected strategy. The 'Debit Put Butterfly' section shows 'Lower' as 736, 'Center' as 741 (highlighted with a blue border), 'W' as 5, 'Upper' as 746, and a lock icon next to a field containing 0.5, with '(Db)' next to it.

Step 8 — Enter the Hedge Price

Type the debit price for the butterfly in the price field and lock it. The BPR updates to include the hedge cost.

XSP Hedge Options

☒ Enable Additional Hedge Qty: BPR: \$60

Hedge: ☐ None ☐ Custom

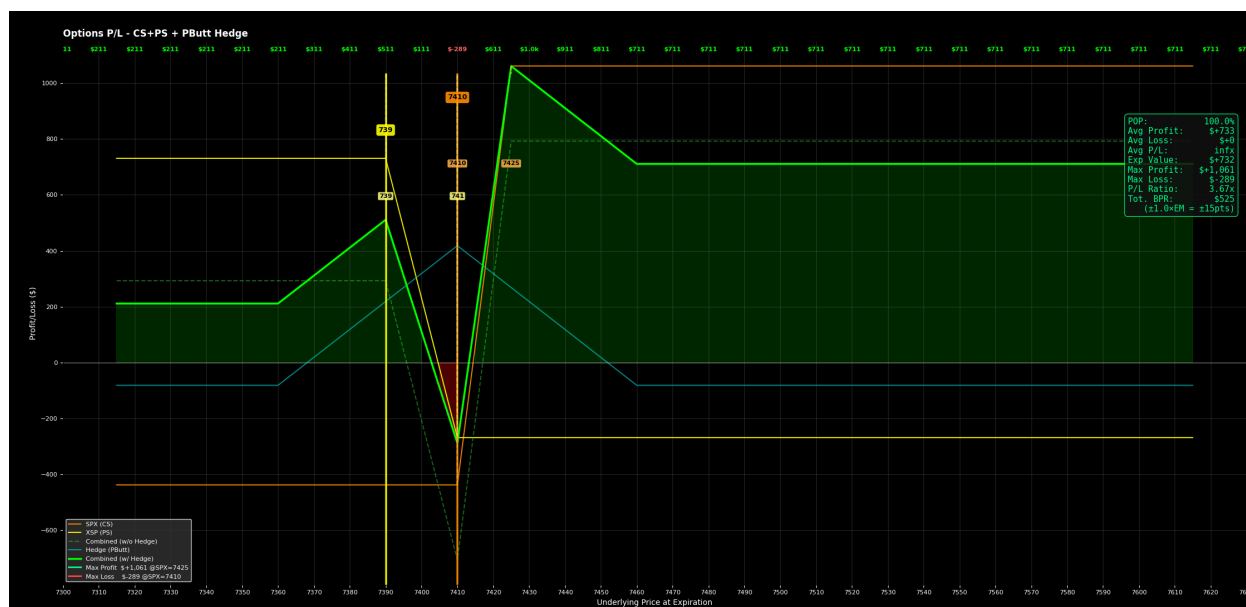
☐ Call ☐ Put ☐ CS ☐ PS ☐ CButt ☒ PButt ☐ IC ☐ CCond ☐ PCond ☐ Straddle ☐ Strangle

Debit Put Butterfly

Lower: Center: W: Upper: (Db)

Step 9 — View the Updated Risk Graph

The chart now shows four curves: SPX CS (orange), XSP PS (yellow), XSP PButt Hedge (blue dashed), Combined without hedge (white dashed), and Combined with hedge (green). Notice how the butterfly hedge creates a profit peak near the center strike, improves Max Profit to \$1,061, and reduces Max Loss to -\$289 compared to the unhedged position. POP rises to 100%.



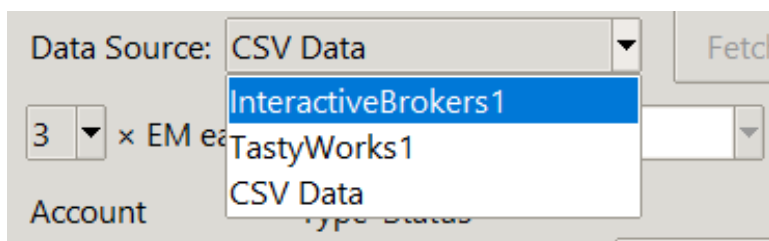
The examples shown are for illustrative purposes only. The actual graph will vary based on real-time prices. The P/L shown is not representative of real results.

Part 2 — Getting Live Prices from Broker

Before proceeding: You will need to add and configure broker accounts for the following steps. For detailed instructions, see the main manual **Section 13.5 — Adding Broker Accounts**.

Step 10 — Select a Data Source

In the main window, click the **Data Source** dropdown. Your registered accounts (InteractiveBrokers1, TastyWorks1) now appear alongside CSV Data. Select the account you want to connect to.



Step 11 — Press Connect

The accounts panel shows each registered broker with its connection status. Click the **Connect** button next to the account you want to use. Make sure TWS or IB Gateway is already running and logged in before connecting.

Account	Type	Status	
InteractiveBrokers1	IB	Not connected	Connect
TastyWorks1	TT	Not connected	Connect

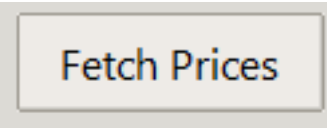
Step 12 — Connection Confirmed

Once connected, the status changes to **Connected** (shown in green) and the Connect button changes to **Disconnect**. The expiration dates available from the broker are loaded automatically into the Expiration dropdown.



Step 13 — Fetch Live Prices

Click the **Fetch Prices** button. The application requests the current mid prices for all active strategy legs from the broker. Price fields update automatically and the chart recalculates. Click again at any time to refresh prices, or the button will switch to **Stop Streaming** if live streaming starts.



Step 14 — Strategies Updated with Live Data

After fetching, all price fields show a **green background** indicating fresh live prices. The chart recalculates using the actual current market prices. In this example SPXW CS shows 6.00 and XSP PS shows 0.01 from the live feed.

Instrument 1 (SPXW)

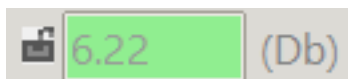
Debit
None
Custom
Call
Put
CS
PS
CButt
PButt
IC
CCond
PCond
Straddle
Strangle
Symbol: SPXW Qty: 1 Mirror 10 BPR: \$600
Call Debit Spread
Long Call: 7560 Width: 15 Short: 7575 6.00 (Db)

Instrument 2 (XSP)

Debit
None
Custom
Call
Put
CS
PS
CButt
PButt
IC
CCond
PCond
Straddle
Strangle
Symbol: XSP Qty: 5 Mirror 1 BPR: \$5
Put Debit Spread
Long Put: 741 Width: 2 Short: 739 0.01 (Db)

Step 15 — Reading a Fresh Price

A **green background** on a price field means the value was just received from the broker and reflects the current market mid price. If the price becomes stale (the market has moved but the price has not been refreshed), the field background turns **red** as a visual warning. Click **Fetch Prices** to refresh. You can lock any field (click the padlock) to hold it at a specific value while other fields continue to update.




Example of Bad Data from CSV File

When loading CSV data, corrupted prices can cause incorrect calculations. Bad data occurs when the calculated spread price does not match the strategy type (e.g., a debit spread shows a credit price or vice versa). This typically happens when the short leg is more expensive than the long leg.

The application detects bad data and displays a yellow background and a warning triangle on the price textbox. Strategies marked with bad data are automatically skipped during optimization.

Bad Data Example:

